Evolutionary Search Comparison with Selected-Frontier Audit Packages

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Draft — May 12, 2026

Abstract

This draft documents the BISECT evolutionary and Pareto comparison path. It separates frontier quality from selected-plan validity and requires fixed-seed metadata, validity-preserving operators, and RPLAN/RCTX audit packages for any selected frontier entry. A non-dominated entry is treated as an observed search result under a declared budget, not as a legal or policy preference by itself.

1 Introduction

U.19 documents `bisect-pareto` and `bisect-cli::pareto_cmd`, exposed through `bisect pareto` and selected-frontier audit flags. The paper is an evolutionary/multi-objective implementation paper, not a proof that a computed frontier is globally complete.

2 Implementation Contract

The implementation contract has three parts: validity-preserving crossover and mutation operators, deterministic fixed-seed frontier metadata, and selected frontier export. Operators may return valid children, explicit valid fallbacks, or rejected-child records; invalid silent outputs falsify the contract.

The atlas view makes the frontier decision explicit. Offspring creation, repair/fallback, frontier observation, and selected-index export are different events. A selected entry need not be best in every objective; it is a chosen trade-off point whose generation, objective vector, validity status, and selection index must travel with the RPLAN/RCTX package.

The implementation therefore separates producing candidates, observing a frontier, and selecting an entry for downstream audit. Frontier membership is a search-output claim. Selected-package status is an audit-readiness claim. Selection-deferred status means the entry can be compared but should not be reported as a chosen plan without the U.14 selection record.

Layer	Record	Claim boundary
Operator-valid	parent ids, seed, parameters, validity status	child validity
Operator-fallback	parent ids, failed operator, fallback reason	validity preservation
Frontier-recorded	objective vectors and dominance metadata	search result under budget
Selected-packaged	chosen entry and audit package id	auditable final plan
Selection-deferred	frontier entry without selection record	comparison only

Table 1: Evolutionary comparison record layers.

3 Audit Artifacts

A selected frontier entry should be exportable as RPLAN/RCTX with certificate and manifest sidecars. The audit package verifies the selected plan and lineage under a declared profile; it does not certify that the frontier is complete or that the selected trade-off is legally preferred.

The manifest must include the fixed seed, population and generation identifiers, operator parameters, parent ids for generated descendants, selected entry id, objective vector, constraints profile, RPLAN path, RCTX path, certificate path, and verification transcript. If a candidate reached the frontier through a fallback or repair path, that path remains part of the selected-entry lineage.

Selection criteria are separate audit inputs. U.19 can package a selected frontier entry for verification, but preference among valid non-dominated entries belongs to the U.14 selection record and the U.15 posture boundary.

The public golden package can be checked either through the RPLAN certificate verifier or through the BISECT manifest verifier.

4 Evaluation Plan

Current evidence includes L0/L1 crossover, mutation, determinism, selected package verification tests, and a public U.19 selected-frontier package in the package corpus. Seed/objective sensitivity, VRA-aware objective disclosure, and frontier-quality comparisons remain future empirical work.

5 Limitations

Search budget affects frontier quality. A valid selected plan does not imply a complete frontier, and a non-dominated entry does not settle policy or legal choice without the U.14 selection record and U.15 posture boundary. Fixed-seed reproducibility is necessary for audit but does not by itself imply fairness, robustness, or legal adequacy.

Claim	Current evidence	Publication gap
Operators preserve validity or emit fallbacks	L0 crossover/mutation tests	Operator transcript examples
Fixed-seed frontier records are reproducible	L0/L1 determinism tests	Seed sensitivity appendix
Selected entries can be audited as RPLAN/RCTX packages	Selected package verification tests plus public manifest and transcript	larger frontier export
Frontier quality is robust across objectives and budgets	Future evidence only	Benchmark and sensitivity protocol
Selected trade-off is legally or policy preferred	Out of scope for U.19	U.14 selection record and U.15 posture analysis

Table 2: Claim and evidence ladder for U.19.

Future empirical work must report budgets, seeds, objective weights or definitions, operator parameters, fallback rates, and sensitivity to VRA-aware inputs before making broader quality claims.

6 Conclusion

U.19 makes evolutionary comparison compatible with the fixed point: exploratory frontiers remain heuristic search objects, while selected outputs become auditable plan packages. The reviewed draft now treats operator validity, frontier observation, selection, and audit packaging as separate layers, preserving room for policy choice without hiding the computational lineage of selected plans.

References

BISECT project. U.19 evolutionary search comparison paper plan, 2026. Local planning document and implementation roadmap.